



## Background and Context

It was a mild February day in Grantshire and the sun was shining through the window of **Alf Handle**'s office in the Highways Maintenance depot.

It seemed strange that he was concerned not about the lovely sunshine but about the ice and snow they had just experienced that winter. He knew he had work ahead for the **next** winter season (2024/25).

Both Alf<sup>1</sup> and **Peter Banbury**<sup>2</sup> – the Chief Executive of the County – were under some pressure to make good decisions regarding the effectiveness of the precautionary road gritting in the area, and manage the re-structuring of the depots without putting the public at risk.

The idea of reducing the depots in Grantshire had been discussed for a few years and now the decision had to be made: the current **three depots** needed to be re-organised into either two or even one, single depot. Alf had been tasked with analysing, planning and costing this significant operational change.

In terms of the weather, the previous winter had been a somewhat strange one – initially it had been quite cold and December had seen a reasonably high number of gritting events (20). However, January turned out to be mild and so the total number of gritting events for this past season was just **38 events**.

In terms of gritsalt, Grantshire's current total capacity across the three depots was **7,000 tonnes**, which had easily been enough to deal with last winter. However, both Alf and Peter were keen to avoid running out of gritsalt in the future, and any changes to the depot structure would need to analyse this issue.

Media coverage of climate change had increased the attention on Peter and tough questions were being asked by politicians and media alike. As an ex-journalist, Peter knew how the media world worked, but he still needed some answers to give them.

Were we sufficiently prepared for an extreme winter?  
Were we gritting the right amount and type of roads?  
Were we using the right quantities of gritsalt?  
Was the stockpile sufficient?  
Would a single depot be better than two depots for dealing with extreme weather conditions?

The next winter 'season' was clearly a long way off but the changes to the depots would need time to complete, so a full gritting analysis was clearly needed.

However, Alf was a busy man and so contacted you for support. On the next page is a copy of the first email from Alf, addressed to you, outlining his needs and concerns.



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Email to: the consultancy team  
From: **Alf Handle**  
Re: 2024/25 Winter Gritting Project

The following outlines your task. Any clarification of scope or requests for further data can be made directly to me, via email.

The main focus of the analysis relates to depot restructuring in Grantshire.

**Depot Restructuring:** The 'green light' has now been given to reduce the depots from three down to either two or one depot. Of the current depots (A, B and C) it has already been decided that depot C will definitely be **closed** and decommissioned this summer, due to it being in poor condition and un-economical to refurbish. This leaves A and B still operational.

Initial discussions have suggested the following possible options:

1. Expand depot A to accommodate the extra vehicles (from depot C), and leave depot B at its current size, or
2. Expand depot B to accommodate the extra vehicles, and leave depot A at its current size, or
3. Expand both A and B to accommodate the extra vehicles, or
4. Expand depot A to form a single, large depot and close depot B altogether, or
5. Expand depot B to form a single, large depot and close depot A, or
6. Close both depots A and B and build a **brand-new depot** elsewhere in Grantshire. There are two possible sites for this new depot, called 'site X' and 'site Z'.

All these options need to house 16 gritting vehicles in total. At this stage, there are no other options being considered, other than the six options given above.

There are two possible sites for the new depot (option 6). The first site – codename 'site X' – is near Tiger Bay, and the second site – 'site Z' – is at the southern tip of Telford Fell (see figure 1.2 in Annex 1). All the options have their strengths and weaknesses and part of your task is to analyse this and document this information, and to make a suggestion as to which site you think would be the best option for Grantshire. All the de-commissioning, construction, changeover and annual depot costs should now be available from **Peter**.

Although any recommendations will focus primarily on costs (operational, construction and changeover), I am also concerned about the **practical** implications too, in terms of gritting delays, vehicle usage and safety, driver conditions and workloads, and practical considerations for depot supply and storage of gritsalt. Note, all costs for the analysis should be based on gritsalt as the de-icing material.

In addition to the depot restructuring there will also be some minor changes to the road network near Lake No, as a new bridge over the lake is near to completion linking Hindlebury and Longtown. The impact of this will be to add some extra roads/km to the gritting schedule.

**Other issues.** The weather, obviously. As you well know, the 2023/24 winter was fairly mild but climate change issues mean we shouldn't assume this to be the 'new normal'. Next winter could well be very different and I suspect the impact of climate change is to make winters more **volatile**, rather than simply warmer (or colder) on average. I oversee all the budgets and I need to be confident that the money we spend on winter gritting is sensible and targeted, and ready for extreme conditions. Any cost analysis should be based

on three types of winter weather: **mild**, **average** and **extreme**, to get a clear picture of the range of costs that might emerge.

**Data and Model.** Dirk Damasko can supply you with a basic costing model in Excel, based on last season. Note that this model is based on three small depots whereas you will obviously have to modify this to work with the new depot structure. The new routes associated with these options will be included in the model. Note, we do not have access to the routing algorithm that generated these routes.

It is unclear at this stage if this model is sophisticated enough to deal with all the issues being faced, and thus may need to be developed further. The Excel file also contains some of the historical data on events and weather in Grantshire, which may be useful.

I have also included three annexes to this brief, describing some important background information on salt, coverage, roads, accidents and accident costs, which may be relevant to your analysis.

**Deliverables.** At this stage, the requirement is that you deliver two things: a client **report** covering all the research issues discussed above, and an in-person **presentation** of your final analysis to myself and Peter Banbury, although this will depend on mine and Peter's availability. The report should contain all the relevant findings and assumptions and be suitable for further distribution to a wider audience, beyond myself and Peter. I suggest around 6 pages will be sufficient, not including a title page and appendices. Note that appendices should be used both sparingly and sensibly.

It may also be sensible to have an interim presentation and/or **progress report** part-way through the consultancy period to update myself and Peter on your progress and findings, and to discuss any assumptions and implications. We can discuss this idea at a later date.

Ultimately, what I want to see is that we are properly considering all the possible scenarios and identifying the most cost/effective plan. We must reassure both Peter Banbury and the public at large that we have analysed this thoroughly. The budget is not 'fixed' but it is important that all options have been costed and thought through properly, and that the people of Grantshire are getting good value and are safe on their roads.

Yours, Alf Handle  
Grantshire Highways Maintenance



Email to: the consultancy team  
From: Peter Banbury  
Re: **Winter Gritting Project**

*I am very pleased to have you on board for this project, as we have some challenges ahead of us, in terms of depot restructuring and enhancing public safety. Your primary contact for the project will be Alf Handle, as he knows the gritting operations extremely well. However, I have a few concerns too.*

*Firstly, I need to know if we are gritting enough roads. Every winter, it seems, we experience the tragic loss of life in ice-related road traffic accidents. Indeed, only last week, a family man from Askham was killed on his way to work when he skidded on ice and hit another vehicle, leaving a wife and young family. Terrible news.*

*Given that we are re-structuring the depots, could we grit more roads at the same time? If so, do you think we should target urban or rural roads? Which are more dangerous? How much would it cost to grit more roads? How much does a life cost, ultimately? We are all under pressure to deliver value but surely we can find a good balance between funding and safety? I will have the final cost estimates for the depot changes very soon so please drop me an email to get those.*

*My goal is to lead Grantshire and invest wisely in its infrastructure and services. Your analysis will contribute directly to this and create a safer Grantshire, and I look forward to seeing it.*

*regards*  
*Peter Banbury*  
*Chief Executive of Grantshire*

## Annex 1

### The Winter Gritting Plan (Last Year)



#### Overview

The main element of winter gritting in Grantshire is the precautionary gritting of roads in anticipation of imminent freezing conditions to stop ice forming on the roadway.

From an operational point of view, this involves having a set of vehicles and drivers that can spring into action very quickly, triggered by weather forecasts that anticipate freezing conditions.

The main elements of such an **‘event’** are:

- Mobilising 16 drivers and their gritsalt-spreading vehicles at the three depots in the County
- Ensuring that the vehicles are loaded with the correct amount of spreading material (gritsalt)
- Following the agreed spreading routes
- Spreading the gritsalt in the appropriate quantities for the chosen roads on these routes
- Returning to the depot to clean the vehicles

In a typical winter, the number of gritting events is around 35 to 60 occasions.

Historical data on events and temperatures is available for the last 59 years.

In terms of timing, a **‘regular event’** sees the vehicles leave the depots at around 6pm and finish their routes by around 9pm.

However, in particularly cold and/or windy situations the gritting is sometimes **repeated** in the early morning, usually around 5am, before most of the public start their journeys to work. This is referred to as a **‘repeat event’**.

#### Planned Routes

There are currently three depots in the County. The total roadway length in the County is **2,143** kilometres but not all of this is gritted. Also, there is the M6 motorway running through the southwest edge of the county but this is the responsibility of the Highways Authority and not the Local Authority – so it is **not** involved in the Alf’s winter maintenance plans.

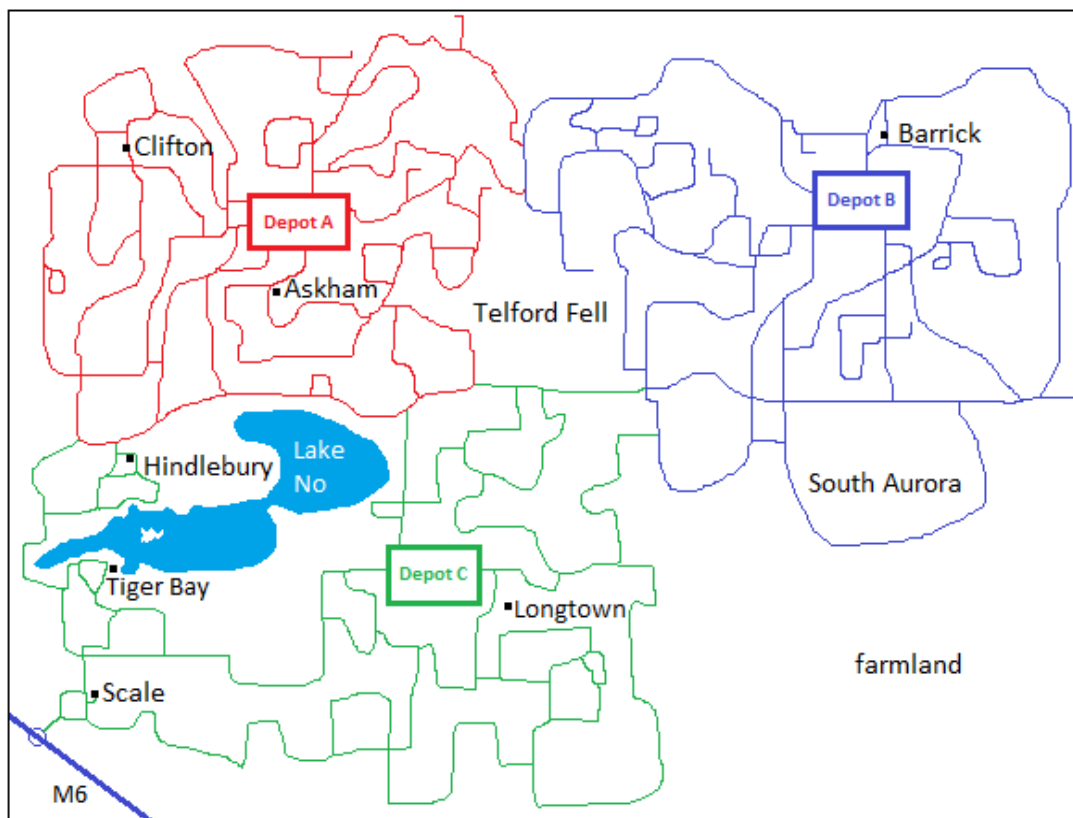
The current gritting routes cover 1,114 kilometres of actual roadway and **1,282** kilometres of **road lanes** (as some roads need to be gritted in both directions).

The routes themselves have been determined using a *routing algorithm*. The algorithm finds the minimum total distance travelled by the 16 vehicles, given the constraints of the problem: the roads that need to be gritted, the capacity of the vehicles, the time available per route, driver workloads and so on.

See figure 1.1 overleaf for a diagram of the 16 routes and three depots.



**Figure 1.1: An illustrative diagram of gritting routes for last season (2023/24)**  
(A: 6 routes, B: 5 routes, C: 5 routes)



The government requires Authorities to report the road coverage percentage of their routes in terms of **road** length gritted divided by total road length – in this case around half (**52%**) of the roads in Grantshire are currently gritted.

No specific targets are set and across England the overall average is 38% with a high of 75% and a low of 20%. The National Audit Office recommends a minimum of 35%. Hence, from this perspective, the coverage in Grantshire is clearly acceptable.

The average lane length gritted per route is around **80** kilometres with a high of 90 kilometres and a low of 67 kilometres.

Within any individual vehicle route there are two types of ‘mileage’; **gritting** mileage and **dead** mileage. ‘Gritting mileage’ is where the vehicle is ejecting gritsalt onto the road and ‘dead mileage’ is when the vehicle is driving but no gritsalt is being ejected.

Dead mileage occurs for three reasons:

1. The vehicle has only just left the depot but hasn’t yet reached the start of its designated gritting route
2. The vehicle is driving on a road that has *already been* gritted previously, by either another vehicle or that same vehicle
3. The vehicle is travelling down a minor road that isn’t designated to be gritted (this is rare though).

The total gritting mileage per event is **1,282** km, and the total dead mileage per event is **368** km, meaning the total mileage (gritting plus dead) covered per event is **1,650** kilometres.

## Gritting Fleet and Gritsalt Usage

Last season the Authority operated 16 gritting vehicles of **10.5 tonnes** (of gritsalt) capacity, except two older vehicles which hold slightly more (11). One very old vehicle is also held in reserve, in case of an emergency breakdown.

The average width of the road lanes gritted in Grantshire is **7.2 metres** and the spreading rate is computer-controlled in terms of grams per metres squared. For precautionary gritting this Authority currently spreads at **15 grams** per square metre, which is somewhat higher than the recommended minimum of 10 grams per square metre *on the roadway* as some material is wasted in the spreading process: for example, it can bounce or be blown into the verges.

During an average winter season, the Authority uses around 6,000 tonnes of gritsalt for precautionary gritting. The capacity of Grantshire's stockpile across 3 depots is around 7,000 tonnes. Should Grantshire **run out** of gritsalt then there is the facility to order more, but the price is always higher than usual during winter due to demand effects (budget for **£35** per tonne).

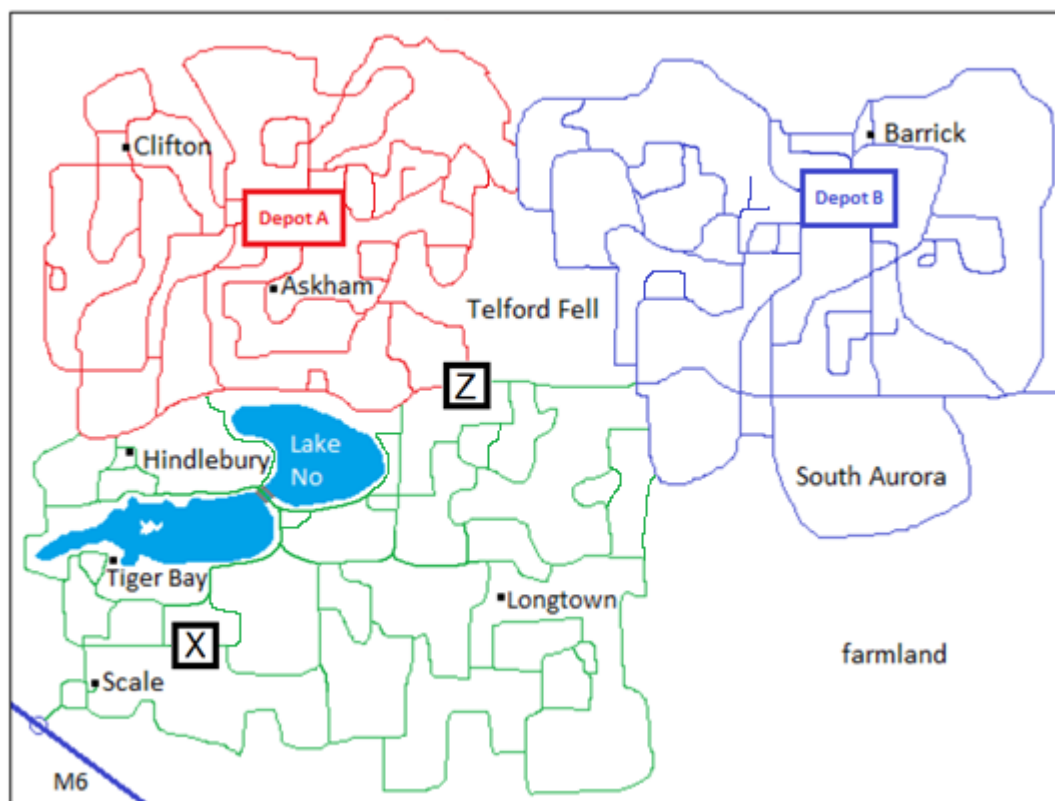
Furthermore, there is a very small chance that national supplies of gritsalt could run out *completely*, thus making re-stocking impossible (or at the least very problematic). This has never happened though.

## Changes for 2024/25

There are **three** changes proposed for next year:

1. Depot restructuring from three depots down to either two depots, or a single depot
3. An increase in the total gritting distance per event, from 1,282 km to 1,333 km, due to new road building
3. A small increase in vehicle speeds, from 40kph to 45 kph

**Figure 1.2: An illustrative diagram of the depots and the new gritting routes for 2024/25**



The gritting routes in Grantshire have been altered slightly to account for additional roads needing to be gritted; primarily a **new bridge** over Lake No (see figure 1.2) plus one or two other roads which have become busier recently. This has resulted in the total gritting mileage ('Lane Length') per event increasing by 51km to **1,333 km**, compared to 1,282 km previously.

## Financial Facts 2023/24

The precautionary gritting budget is part of a larger 'Winter Maintenance' budget.

Alf Handle has produced a basic of the cost breakdown for the precautionary gritting activity last year, shown in Table 1.1. with more detailed breakdown shown in Table 1.2.

**Table 1.1: The Costing for Precautionary Winter Gritting for 2023/24**

|                          |                 |
|--------------------------|-----------------|
| Cost of Gritsalt         | £131,000        |
| Vehicle Running Costs    | £25,000         |
| Annualised Vehicle Costs | £80,000         |
| Driver Wage Costs        | £49,000         |
| Depot Costs <sup>3</sup> | £36,000         |
| <b>Total</b>             | <b>£321,000</b> |

These numbers are approximate because it requires an apportionment of costs not normally required by the financial accounting procedures within the Highways Department. The values are lower than usual as the gritting activity last year was below average.

Some of the main cost parameters used by Alf in reaching his conclusions are:

|                                     |                                 |
|-------------------------------------|---------------------------------|
| Purchase cost per tonne of gritsalt | £25                             |
| Vehicle Running Cost                | £0.40 per km                    |
| Vehicle Purchase Cost               | £80,000                         |
| Life of Vehicle (assumed)           | 16 years                        |
| Operative Wage Cost                 | £20 per direct operational hour |

Most of these costs are effectively stable year-on-year. The only cost which may fluctuate is the price of gritsalt, but this is expected to be stable at £25 for the near future.

## Other Comments on the Operation

- The precautionary activity is usually instigated in the evening when the Met Office forecasts freezing temperatures during the night. This forecast is usually made mid-afternoon.
- When a gritting event is initiated, **all 16 vehicles** from all three depots are deployed at 7pm. There is no 'partial gritting' of any sort whatsoever.
- It is policy that the gritting itself should be completed within **3 hours** of being instigated.
- However, all 16 drivers are paid for **4 hours** work on a gritting event, as this includes 1 hour of pre- and post-gritting activities, such as loading the vehicle and washing-down the vehicles afterwards.
- In very unusual circumstances (weather, delays etc.) the drivers may have to work for more than 4 hours. If so, they are paid 'double-time' for the extra time, so **£40** per hour, paid in whole hours.
- The vehicles last year ran at an average gritting speed of **40 km** per hour. However, this value will be increased to **45kph** for 2024/25, for both gritting and dead mileage.
- Sometimes a precautionary activity will be **repeated** in the early morning, at around 5am, in situations when it has been severely cold and/or very windy during the night. This is called a **repeat event**.

<sup>3</sup> This is the 23/24 figure, summed across all three depots. However, the depot costs for next year will obviously change, as the depots are restructured. These new depot costs should be available soon.



**Table 1.2      Gritting Information for Grantshire**

| <b>Term</b>                        | <b>2023/24<br/>value</b> | <b>Units</b>                                                                                                                                           | <b>Notes</b>                                                                                                                               |
|------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Number of Events                   | 38                       |                                                                                                                                                        | Number of separate times the gritting vehicles were deployed last year                                                                     |
| Number of Gritting vehicles/routes | 16                       |                                                                                                                                                        | The number of vehicles used. This is equivalent to the number of separate gritting routes.                                                 |
| Spread rate                        | 15                       | grams/square metre                                                                                                                                     | the amount of gritsalt ejected by the gritting vehicle [equivalent to approx. 1200 grams per second]                                       |
| Lane width                         | 7.19                     | metres                                                                                                                                                 | Width of the road, on average                                                                                                              |
| Spread per km                      | 107,850                  | grams/km                                                                                                                                               | $15 \times 7.19 \times 1000\text{m}$                                                                                                       |
| Tonnes per km                      | 0.10785                  | tonnes/km                                                                                                                                              | $107,850 \div 1,000,000$                                                                                                                   |
| Price of gritsalt                  | £25                      | per tonne                                                                                                                                              | current price                                                                                                                              |
| Current lane length                | 1282                     | kilometres                                                                                                                                             | This is the length of road lanes gritted                                                                                                   |
| Roadway length                     | 1114                     | kilometres                                                                                                                                             | The length of the roadway gritted. This is slightly shorter than the lane length, as some sections of road are gritted in both directions. |
| Distance per Event                 | 1650                     | kilometres                                                                                                                                             | Total distance driven by the vehicles per gritting event                                                                                   |
| Vehicle capacity                   | 10.5                     | tonnes                                                                                                                                                 | The capacity of the vehicles (two are slightly bigger)                                                                                     |
| Average speed                      | 40                       | km per hour                                                                                                                                            | Speed when gritting (this will increase next year)                                                                                         |
| Vehicle running cost               | £0.40                    | £ per km driven                                                                                                                                        | Including fuel, maintenance and all servicing                                                                                              |
| Wage cost                          | £20                      | £ per hour                                                                                                                                             | For each driver                                                                                                                            |
| Paid hours per event               | 4                        | hours                                                                                                                                                  | Each driver is paid for 4 hours' work                                                                                                      |
| Vehicle cost (per vehicle)         | £5,000                   | For the purposes of the accounting, each vehicle costs £80,000 to buy and is assumed to last 16 years, so this works out at £5,000 per operating year. |                                                                                                                                            |
| Management cost                    | £36,000                  | £                                                                                                                                                      | Annual cost of running the three depots                                                                                                    |
| Total gritsalt spread              | 5,254                    | tonnes                                                                                                                                                 |                                                                                                                                            |

## **Annex 2**

### **The Properties of Salt and Other Materials for Winter Gritting**

**John Thornton**

**June 2019**

Highway salt – also referred to as *gritsalt* or *rock salt* – is used in large quantities around the world to keep roads clear of ice and snow.

In the United States approximately 15 million tonnes of salt are spread each winter – a figure that has increased from an average of 10 million tonnes in the 1980s.

Salt is relatively cheap to buy (\$40 per tonne in the US and £20-25 per tonne in the UK) and other potential de-icing chemicals are much more expensive. Vitaliano (1992)<sup>4</sup> looked at the use of Calcium Magnesium Acetate (CMA) in the US as an alternative which was priced at \$650 per tonne at that time.

Although salt is relatively inexpensive, readily available and effective, it does have the unfortunate side effects of being environmentally unfriendly, and is damaging to both metallic infrastructure due to corrosion and concrete due to spalling.

However, substantial capital has been invested in developing a salt supply, distribution and deployment infrastructure. Therefore if additives are available, at a reasonable cost that militates against such deleterious effects the resultant product may be seen as holistically more cost-effective and operationally elegant as an effective winter maintenance material.

In England gritsalt sales in very cold winters can reach 2 million tonnes but the average is closer to one million tonnes per winter.

Gritsalt is normally composed of at least 90% sodium chloride, 2% to 4% water and the remaining constituents are insoluble particles that give extra traction and anti-caking agents to keep the salt friable and easy to spread. Gritsalt is normally spread as 6 mm or 10 mm particles.

Although there have been other chemicals used on the UK road network, for example Urea on the Midland Links elevated sections, gritsalt has completely dominated the market since it was introduced in the early 1960s. Most other countries in Europe use wetted salt or brine but this has never caught on in the UK as 'dry' gritsalt is normally able to go into solution when spread, due to the high humidity in the UK's maritime climate. In drier climates, getting the salt to go into solution can be a problem.

### **Corrosion Costs**

#### **Vehicle Corrosion**

Today most new cars are supplied in the UK with extended (6 to 10 years) corrosion warranty guarantees. This covers any corrosion to the car bodywork but does not normally cover corrosion to car exhausts.

Vehicle corrosion due to salt has been estimated by TRL at around £200 million per annum in England which is the equivalent of £8 per annum of corrosion for each of the 25 million vehicles due to salt. It also equates to approximately £200 per annum per tonne of salt spread.

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<sup>4</sup> Vitaliano, D.F. (1992) An Economic Assessment of the Social Costs of Highway Salting and the Efficiency of Substituting a New De-icing Material, *Journal of Policy Analysis and Management*, 11(3), 397-418

## **Annex 3**

### **Accident Benefits of Precautionary Salting**

**John Thornton**

**June 2019**

#### **Background**

One of the main areas of benefit of planned winter road salting is that it reduces road accidents and the costs associated with them.

There were 2,782 fatal accidents, 24,873 serious injury accidents and 159,792 slight injury accidents on England's roads in 2019. Table 3.1 shows the number of these accidents that occurred on Highways Agency roads – motorways and trunk roads – principal A roads and rural and urban minor roads.

**Table 3.1: Accidents by Type of Road**

| Road Type       | Fatal | Serious | Slight  | Total   |
|-----------------|-------|---------|---------|---------|
| Highways Agency | 487   | 2,515   | 21,441  | 24,443  |
| Principal A     | 1227  | 9,834   | 58,304  | 69,365  |
| Minor (Urban)   | 708   | 8,933   | 64,087  | 73,728  |
| Minor (Rural)   | 360   | 3,591   | 15,960  | 19,911  |
| Totals:         | 2,782 | 24,873  | 159,792 | 187,447 |

Expressing accident rates in terms of traffic volumes – billion vehicle kilometres – the relative rates are shown in Table 3.2.

**Table 3.2: Accident Rates per Billion Vehicle Kilometres**

| Road Type       | Fatal | Serious | Slight | BVKM |
|-----------------|-------|---------|--------|------|
| Highways Agency | 3.2   | 16.8    | 142.9  | 150  |
| Principal A     | 10.9  | 87.9    | 520.6  | 112  |
| Minor (Urban)   | 6.7   | 85.2    | 610.3  | 105  |
| Minor (Rural)   | 9.5   | 94.5    | 420    | 38   |

It can be seen that Local Authority roads are significantly more dangerous than motorways and trunk roads per vehicle kilometre with A roads and Minor roads having similar characteristics – although A roads have a higher fatal accident rate.

With respect to rural roads, these are more dangerous than urban roads (per vehicle kilometre) in terms fatal and serious accidents, as the average speeds tend to be higher on rural roads, and thus the severity of the accident will tend to be higher.

## Accidents on Roads that are not subject to Precautionary Salting

It is not possible to say with certainty how accidents rates compare on roads that are subject to planned precautionary salting in relation to those roads not so covered because road accident statistics have not been collected in this way.

For England as a whole the overall coverage of the winter gritting programme in terms of the proportion of Local Authority road kilometres covered is around 38% and, hence, **62%** are unsalted. Since the principal-A roads (always salted) account for around 9% of the total road length, on average, only **29%** (38% minus 9%) of the minor roads are covered by salting programmes, and therefore 71% of the minor roads are not salted.

Thus – *if accidents occur in the same ratio* – we can say that up to 71% of the minor road accidents (71% of 93,639 = 66,484) *could* have occurred on roads that are not subject to precautionary salting.

An unpublished survey of ten Local Authorities carried out by the Audit Commission in their review of coverage rates found that in this sample around 9% of the *rural* minor roads received precautionary salting but that around 35% of the *urban* minor roads were covered. This result suggests that an *urban* minor road is around **3.89** times more likely to receive salting treatment than a *rural* minor road.

In England as a whole, there are 101,414 km of *urban* minor roadway and 86,946 km of *rural* minor roadway: a total of 188,360 kilometres. By applying the 29% coverage of minor roads we obtain an estimate of **54,624** km of minor roads are salted. And thus 133,736km of unsalted minor roads.

Thus,

$$\begin{array}{lcl} \text{rural minor roads salted (km)} & + & \text{urban minor roads salted (km)} \\ \text{or, } (p \times 86,946\text{km}) & + & (3.89 \times p \times 101,414\text{km}) \end{array} = 54,624 \text{ km}$$

where  $p$  = proportional coverage of rural minor roads

$$\begin{aligned} \text{thus, } p &= \frac{54,624}{(3.89 \times 101,414) + 86,946} \\ p &= \frac{54,624}{481,446} \\ p &= 0.11346 \quad \text{or } \mathbf{11.35 \%} \end{aligned}$$

In other words, 11.35 % of *rural* minor roads are salted and  
 $3.89 \times 11.35\% = 44.13\%$  of *urban* minor roads are salted.

Subsequently,  $1 - 11.35\% = 88.65\%$  of *rural* minor roads are **unsalted**  
and,  $1 - 44.1\% = 55.87\%$  of *urban* minor roads are **unsalted**

Using the data shown in Table 3.1, this suggests:

$$\begin{array}{lll} \text{Estimated number of accidents on unsalted } \textit{rural} \text{ minor roads} & = 19,911 \times 88.65\% & = 17,651 \\ \text{Estimated number of accidents on unsalted } \textit{urban} \text{ minor roads} & = 73,728 \times 55.87\% & = 41,189 \end{array}$$

A total accident rate on unsalted minor roads of  $17,651 + 41,189 = \mathbf{58,841}$  per annum  
This around 31% of the total accidents (187,447).

Of course, only a relatively small proportion of these accidents will have occurred during adverse icy conditions and in only a proportion of these will the road condition have been a contributing factor. An important question is that of estimating these proportions.

## The Contribution of Icy Conditions to Accidents

- A German Study

The only direct study of the effects of road gritting on accidents on rural roads has been in Germany carried out by the Darmstadt Technical University (OECD 1989)<sup>5</sup> and this work concluded that the accident rate in the four hours before salting was 3.65 times the rate following salting. This study further concluded that in relation to rural roads in Germany around **15 %** of all accidents are directly affected by whether or not the roads involved are included in a planned salting programme.

- Official Statistics on Road Conditions and Accidents

The STATS19 data in England (Department for Transport) does include information intended to reflect the contributions of road conditions to accidents. However there are significant difficulties in interpreting these data.

The data from the STATS19 database suggests that a maximum of around 8% of all reported accidents might be directly influenced by 'slippery' road conditions.

This would imply that 223 fatalities, 1,990 serious injury accidents and 12,783 slight injury accidents (totalling 14,996) in the survey year (2019) in England could have been so affected.

If the assumption were to be made that all such accidents are on unsalted roads, this would translate into a **25.5%** effect on the unsalted roads ( $14,996 \div 58,841$ ). However, not all 'slippery' accidents are the result of ice, not all reported 'slippery' conditions necessarily contributed to the actual accident and not all such accidents occurred on roads not subject to precautionary salting.

- A Conservative Estimate

Overall the German result is perhaps the most robust basis for assessing the accident reduction effects of salting since it was a direct study of the effect. This German study estimated that 15% of all accidents are affected by the winter gritting activity whereas the STATS19 data suggests a maximum of 25.5%.

In the cost benefit analysis in the next section the more conservative estimate of 10% has been applied. Given the complex nature of many accidents, with many contributing factors, a value of 10% is seen as a reasonable value to use.

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<sup>5</sup> Winter Maintenance and Road Safety, Road Transport Research, OECD, Paris, 1989, pp 107-113

## The Costs of Preventable Accidents

**Table 3.3: Official Department for Transport Accident Costs (per accident)**

| Category       | Cost       |
|----------------|------------|
| Fatal          | £1,447,490 |
| Serious Injury | £168,260   |
| Slight Injury  | £16,750    |
| Damage Only    | £1,490     |

Multiplying these costs by the number of accidents described in table 3.1 produces an annual estimate of around £10.9 billion for all the injury accidents in England, for all types of road, including motorways.

In addition, there is a cost of around £3.35 billion for 'damage-only' accidents, based on 2,060,371 damage-only accidents on principal and minor roads, and 185,767 on motorways.

Thus the total accident cost for all roads (injury and damage-only) is around £14.235 billion.

The total number of injury accidents is 187,447 and the total damage-only accidents is 2,246,137.

Thus the total number of accidents (injury and damage-only) = 2,433,584

Hence, the overall average accident cost is:  $£14,235,308,826 \div 2,433,584 = \text{£}5,850$  per accident

In terms of **minor roads** (and particularly the unsalted minor roads) the picture is slightly different though. Accident numbers from table 3.1:

| Road Type               | Fatal | Serious | Slight | Total  |
|-------------------------|-------|---------|--------|--------|
| Minor (Urban & Rural)   | 1068  | 12,524  | 80,047 | 93,639 |
| Percentage of each type | 1.14% | 13.37%  | 85.48% | 100%   |

Using these percentages, we can estimate the number of each type of accident on unsalted minor roads, based on the total of 58,841 accidents given previously.

This produces the following:

| Road Type               | Fatal | Serious | Slight | Total  |
|-------------------------|-------|---------|--------|--------|
| Minor (Urban & Rural)   | 671   | 7,870   | 50,300 | 58,841 |
| Percentage of each type | 1.14% | 13.37%  | 85.48% | 100%   |

Multiplying these numbers by the accident costs in Table 3.3 produces a total cost of all injury accidents of £3.138 billion.

Data suggests that damage-only accidents are 12.64 times more numerous than injury accidents, hence the number of damage-only accidents on unsalted roads is:  $12.64 \times 58,841 = 743,748$

Multiplying this number by the damage-only cost of £1,490 given in table 3.3 produces a damage-only cost on unsalted roads of £1.108 billion.

Summing the injury and damage-only costs produces an estimated total accident cost on unsalted minor roads of £4.246 billion, annually (£4,246,308,759 to be precise)

Using the conservative assumption of a 10% effect of winter gritting programmes this gives an estimate of around **£425 million** of potentially realisable extra benefits from additional road coverage of the precautionary salting programme.

Given we have 133,736km of unsalted minor roads, given previously, this suggests that if we salted all roads we would see a benefit of around **£3,175** per for every (extra) kilometre salted.